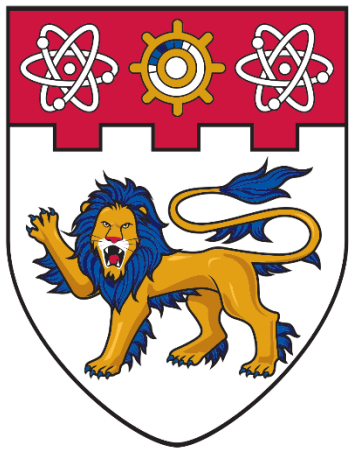




CN Yang Scholars Programme

Final Year Project Title:

Spatial Interpolation Under Varying Uncertainty – Comparison of Methods

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Motivation

Spatial interpolation (SI) plays a crucial role in environmental sciences to predict a value of a variable at points within the same region of sampled locations. The results of SI depend on the choice of spatial interpolation methods (SIMs) and other factors such as data nature, etc. It is interesting to explore how different SIMs account for data uncertainty differently, resulting in varying interpolation results.

Methodology

Dataset Generation

Simulated dataset of earthquake intensity residual (Fig 1) is generated using two data generating methods (DGM): ISK and MVN. The case study residual intensity data of the 2019 Ridgecrest earthquake is obtained from the USGS “Did You Feel It?” (DYFI) intensity map (Fig 2). The *std* associated to each value of number of responses (*nresp*) in a DYFI block follows the empirical equation (1) (Worden et al., 2012).

Invariant-sill Kriging (ISK)

We normalise the intensity residual by multiplying it with their associated *std* values and conducting simple kriging (equation (2)) on rescaled residuals

Multivariate Normal (MVN)

MVN introduces an additional uncertainty in an additive way, resulting in a modification of covariance matrix Σ through an adjustment factor. Equation (3) is the final equations for the MVN.

$$\text{Equation}$$

$$\text{std} = 0.09 + 0.25 \exp\left(\frac{-nresp}{24.02}\right) \quad (1)$$

$$\hat{Y}(\vec{u}_0) = E[Y(\vec{u}_0)] \left(1 - \sum_{i=1}^n \lambda_i\right) + \sum_{i=1}^n (\lambda_i Y(\vec{u}_i)) \quad (2)$$

$$\mu_{Y_1|Y_2} = \mu_{Y_1} + \Sigma'_{Y_1 Y_2} \Sigma_{Y_2 Y_2}^{-1} \zeta' \quad (3)$$

$$\mu_{Y_1|Y_2} = \mu_{Y_1} + \Sigma'_{Y_1 Y_2} \Sigma_{Y_2 Y_2}^{-1} \zeta'$$

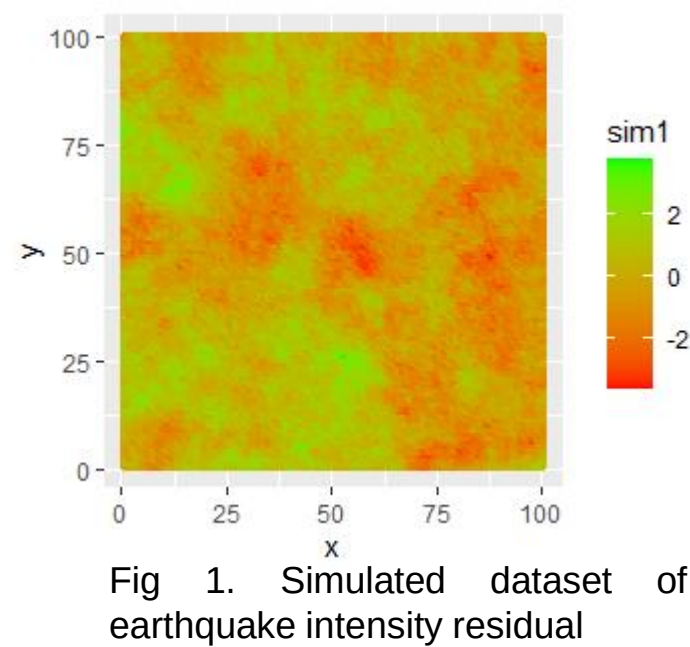


Fig 1. Simulated dataset of earthquake intensity residual

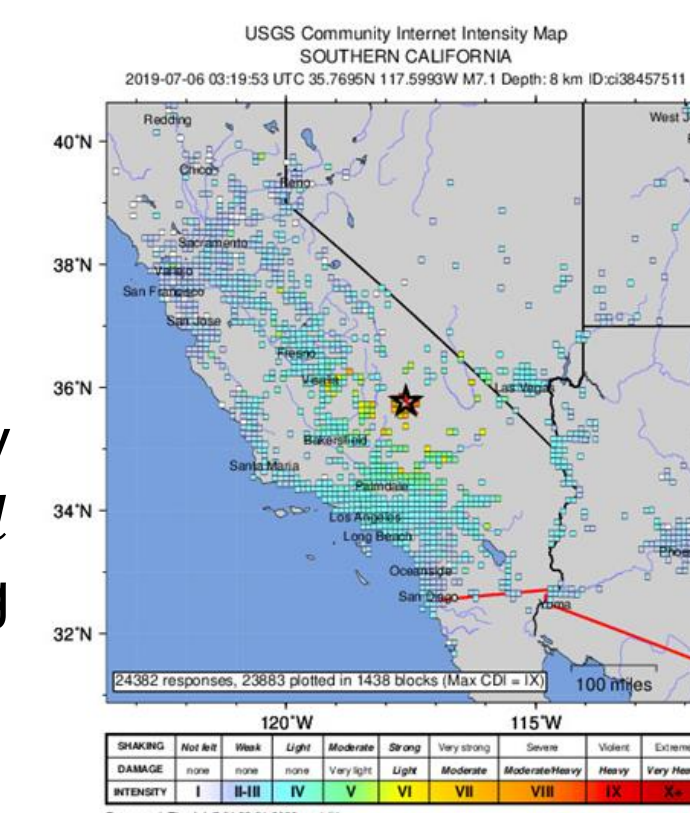


Fig 2. Map of DYFI intensity observations for the 2019 Ridgecrest earthquake

Y	: residual intensity
$\vec{u}$	: location
λ_0	: regional mean of residual intensity
λ_i	: kriging weight at location $\vec{u}_i$
μ	: mean of residual intensity
Y_1	: theoretical intensity based on ground motion prediction equation (GMPE)
Y_2	: observed DYFI intensity
ζ	: DYFI residual intensity
Σ	: covariance matrix

Objectives

This project aims to explore and compare how different spatial interpolation methods (SIMs) account for data uncertainty, resulting in varying spatial interpolation values in the case of predicting earthquake intensity. This study focuses on performing the invariant-sill kriging (ISK) and the conditional multivariate normal (MVN) introduced by Worden et al., 2018 on a simulated earthquake intensity data the 2019 Ridgecrest earthquake data.

Results & Analysis

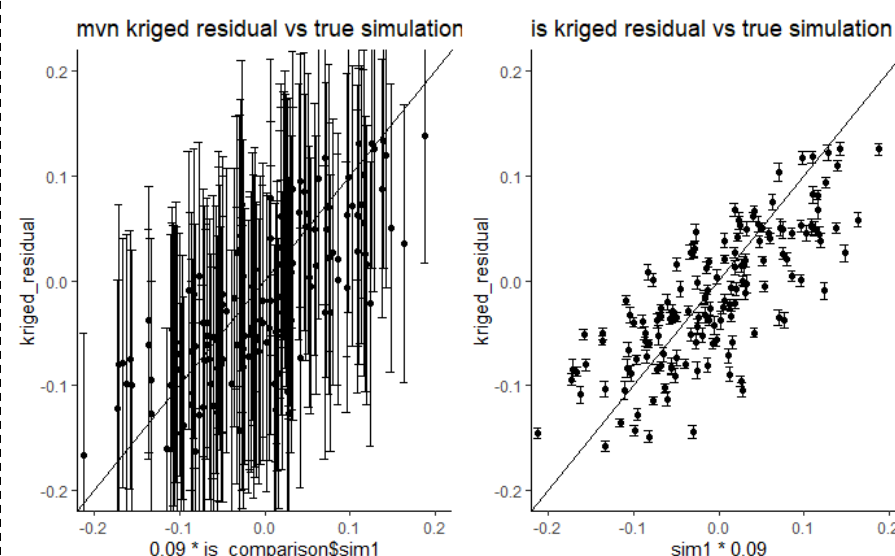


Fig 4. Scatter plot of prediction error: interpolated residual vs true simulation for ISK DGM dataset fitted to MVN (left) & ISK (right) SIM models.

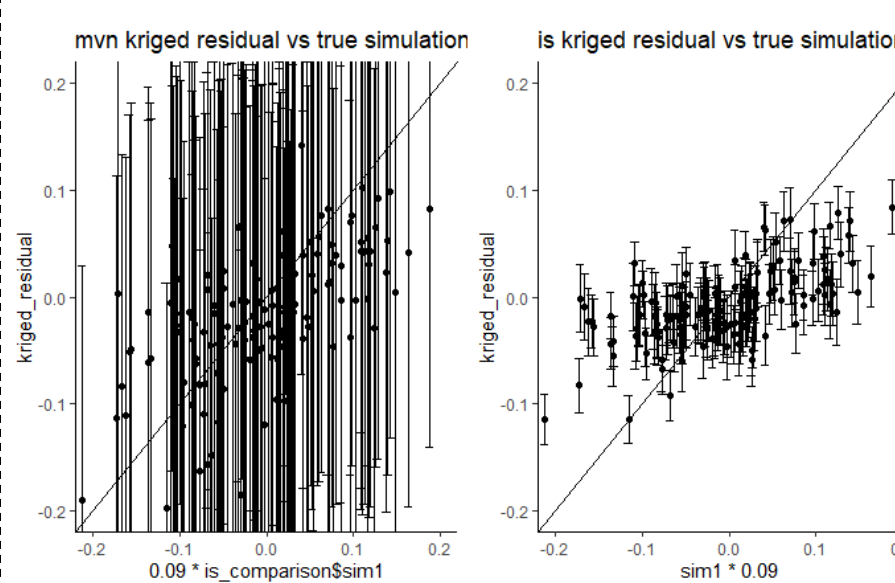


Fig 5. Scatter plot of prediction error: interpolated residual vs true simulation for MVN DGM dataset fitted to MVN (left) & ISK (right) SIM models.

Simulated Data by ISK DGM

When data doesn't have too much noise, ISK gives better interpolation (lower MSE) and fits the data better. MVN estimates have wider prediction variance.

Simulated Data by MVN DGM

When data has a huge noise-signal ratio, MVN, the more robust method, fits and can account for the varying uncertainty ϵ of the data better (lower MSE). However, ISK IQR is still smaller than that of MVN.

ISK vs MVN on the 2019 Ridgecrest Data

No significant observable interpolation difference between ISK & MVN in Fig 5. However, ISK model yields better predictive scores (Table 1).

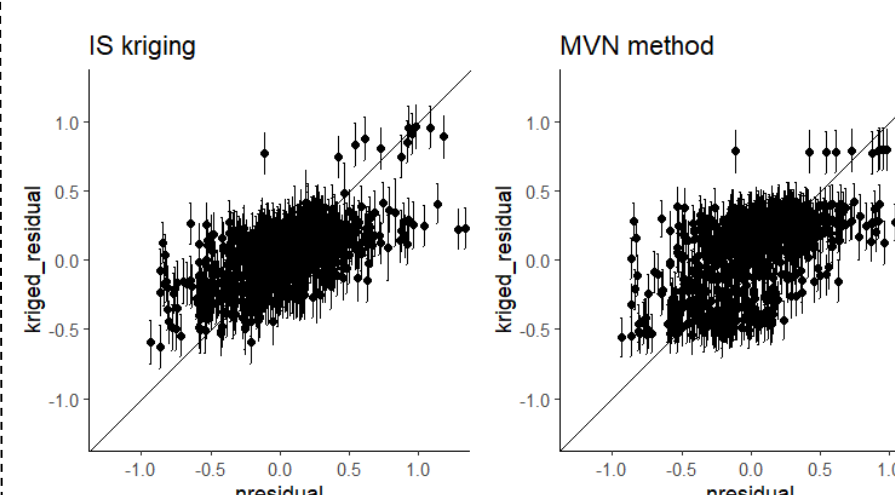


Fig 6. Scatter plot of prediction error: interpolated residual vs observation (nresidual) when the 2019 Ridgecrest residual data is fitted to ISK and MVN models

Table 1. Comparison of predictive scores between the ISK (left) and the MVN (right) models fit the 2019 Ridgecrest earthquake residual intensity data

Predictive Score	Invariant-sill Kriging (ISK)	Multivariate Normal (MVN)
MAE	0.22302	0.42003
MSE	0.08222	0.29675
LogS	0.16957	7.5365
CRPS	0.15979	0.82513

Conclusion

- When fitting simulated earthquake residual intensity data, there is an inconclusive result on whether the the invariant-sill kriging (ISK) and the conditional multivariate normal (MVN) method performs better and give more accurate spatial interpolation.
- When fitting a real-world earthquake intensity dataset from the 2019 Ridgecrest earthquake, ISK performs better than MVN.

References

- Worden, C. B., Gerstenberger, M. C., Rhoades, D. A., & Wald, D. J. (2012). Probabilistic Relationships between Ground-Motion Parameters and Modified Mercalli Intensity in California. *Bulletin of the Seismological Society of America*, 102(1), 204–221. <https://doi.org/10.1785/0120110156>
- Worden, C. B., Thompson, E. M., Baker, J. W., Bradley, B. A., Luco, N., & Wald, D. J. (2018). Spatial and Spectral Interpolation of Ground-Motion Intensity Measure Observations. *Bulletin of the Seismological Society of America*, 108(2), 866–875. <https://doi.org/10.1785/0120170201>